

PRITHA MISHRA

AI Engineer | [Portfolio](#)

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ABOUT

AI Engineer with production experience building and shipping GenAI systems with agentic pipelines, multi-LLM scoring engines, and MCP servers for private equity and energy clients. Deeply hands-on with LLM orchestration, RAG, and evaluation frameworks; driven by building systems reliable and scalable enough to run in the real world.

SKILLS

AI Engineering: LLM Orchestration, RAG Pipelines, Agentic Pipelines, LLM-as-Judge Evaluation, Prompt Engineering, MCP Server Development, FastMCP, Gemini API, Azure OpenAI

ML & Computer Vision: Computer Vision, Anomaly Detection (Isolation Forest, One-Class SVM), PyTorch, Scikit-learn, OpenCV, pandas, NumPy

Full-Stack Development: Python, Flask, FastAPI, HTML, JavaScript, Asyncio, REST APIs

Cloud & DevOps: Azure Functions, Azure Monitor, Azure DevOps, OpenTelemetry, Git

Databases: MongoDB, Cosmos DB

WORK EXPERIENCE

AI Engineer Intern – Quantum Capital Group, USA (Tiger Analytics)

Jan 2026 – Present

Private Equity & Energy Domain

Investor Prospect Engine

- Engineered a GenAI-powered LP screening system across **~7,000 institutional prospects**, fusing DealCloud, Preqin, PitchBook, Excel, and Fintnx.
- Designed a multi-LLM scoring framework with Gemini 2.5 Flash + GPT-4o in parallel via asyncio constituting a 3-layer signal logic (energy fund history, ticket size $\geq$ \$25M, ESG override) and majority-vote consensus.
- Delivered a production Flask + JS SPA **used daily by the fundraising team**, cutting prospect review time from hours to seconds.

Zero-Touch Agentic LLM Evaluation Pipeline

- Replaced a **manual eval workflow** with a fully automated agentic pipeline that is triggered end-to-end via a single chat phrase.
- Built a FastMCP server (4 orchestration tools) to fetch ground-truth from Cosmos DB, invoke the **production QFI RAG pipeline**, and persist results with atomic auto-incremented test case IDs.
- Integrated with Gemini 2.5 Flash LLM-as-Judge scoring pipeline, automated and enabling repeatable, consistent evaluation runs.

Academic Research Intern – National University of Singapore (NUS)

Jun 2025 – Jul 2025

Asia's top University

Insider Threat Detection

- Led a 5-member team building an AI-based Insider Threat Detection system on **100,000+ enterprise logs**; ensemble anomaly detection (Isolation Forest + One-Class SVM) achieved **18% precision improvement** over baselines.

PROJECTS

AI-Powered Robotic System for Hospital Waste Segregation

- Built an end-to-end CV pipeline with custom CNN/SVM model on **~800 labeled biomedical waste images** that integrated with real-time OpenCV inference and GPIO-controlled 6-DOF robotic arm.
- Achieved **~92% classification accuracy and 95% sorting precision**, minimizing direct human contact with hazardous waste.

ReFaceIt – Face Sketch-to-Image Translation

- Trained a Pix2Pix conditional GAN on CUHK Face Dataset to synthesize photorealistic images from hand-drawn sketches, achieving **SSIM > 0.88**.
- Implemented training stabilization techniques (adaptive dropout, LR decay) to mitigate mode collapse.

PUBLICATIONS

Vision-Based Teleoperation of a Humanoid Robotic Arm Using Real-Time Hand Gesture Mapping

International Journal of Engineering Trends and Technology | **Under Review**

CERTIFICATIONS

- Applied Machine Learning with Generative AI — National University of Singapore
- Big Data Analysis and Deep Learning — National University of Singapore
- Integrating AI with Mechanical Engineering — IIT Guwahati

EDUCATION

B.Tech – Electronics and Instrumentation Engineering – SRM Institute of Science and Technology

Jun 2022 – Jun 2026

SGPA: 10 | CGPA: 8.1

RECOMMENDATION

Industry Recommendation from Chief AI Officer & Managing Director | Quantum Capital Group

Academic Recommendation from Head of Department | SRM Institute of Science and Technology